

Aniruddh Nukal

Brendan Santoianni

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Solution of Few Body Problems using Stochastic Variational Method (SVM)

Solution of Few-Body Problems with the Stochastic Variational Method, I. Central Forces with Zero Orbital Momentum by K. Varga and Y. Suzuki presents Fortran code which attempts to numerically solve few-body problems with the Stochastic Variational Method (SVM). The SVM is considered an automatic and universal solution for the purposes of the problem. The code produced by Varga and Suzuki is general with an arbitrary N number of particles in addition to applicability in a number of systems (Coulombic, quark, or nuclear) with a variety of potentials (Coulomb, Yukawa, Gaussian, etc). The code produced is bounded by the memory and computing capabilities of the operating computer in addition to the natural progression from accurate solutions at low N systems to approximate upper bounds at high N systems.

In the consideration of an N-body system with arbitrary values of particle i of mass m_i , spin s_i , isospin t_i , charge z_i , and position r_i ; the many-body Schrodinger equation is the target to solve. Basis functions of the system are assumed to have the form in (2).

$$\begin{aligned}\psi_{SM_s TM_t}(\mathbf{x}, A) &= \mathcal{A}\{G_A(\mathbf{x})\chi_{SM_s}\eta_{TM_t}\}, \\ G_A(\mathbf{x}) &= e^{-\mathbf{x}A\mathbf{x}/2},\end{aligned}\tag{2}$$

Within these basis functions, the antisymmetrizer, spin, and isospin components are represented (although isospin components are neglected in a Coulombic system). A, within this context, is an (N-1)x(N-1) symmetric positive definite matrix, whose diagonal elements correspond to nonlinear elements of a Gaussian expansion while the non-diagonal elements

correspond to interactions between particles. This setup is assumed to have an orbital angular momentum $L=0$. The wave function is expanded in (3).

$$\Psi = \sum_{i=1}^K c_i \psi_{SM_S TM_T}(\mathbf{x}, A_i) , \quad (3)$$

An upper bound for ground state energy is found by solving for the generalized eigenvalue problem of (4).

$$HC = E_K BC , \quad (4)$$

The lowest value eigenvalue is therefore chosen to serve as that upper bound. In the solving of the generalized eigenvalue problem, the nonlinear entries of A (along the diagonal) must be chosen adequately to optimize the accuracy and precision of the SVM. The SVM proposes that these parameters are found by first randomly generating sets of parameters for A. By proceeding to solve the generalized eigenvalue problem for the lowest ground state energy eigenvalue, the parameter corresponding to the lowest state is chosen as a basis parameter. This basis parameter is added to the previous basis states with the basis dimension incrementing by one.

Through the usage of the correlated Gaussian basis, the Hamiltonian can be separated into matrix elements to find the kinetic energy and potential separately. The matrix element of the kinetic energy can be written in terms of A, Λ (a diagonal matrix of reduced masses), and the correlated Gaussian. Using Jacobi coordinates, the matrix element of the potential can then be found by multiplying the correlated Gaussian by an integral which includes the potential's radial form $V(r)$.

In the implementation of the SVM in Fortran, the basis state parameters are generated in three distinct methods:

- (1) Executing the calculations on a predefined basis input into a separate file and then read by the code.
- (2) Parameters are selected randomly and then the steps described in the selection of basis parameters for A above, increasing the basis dimension incrementally.
- (3) Initial values for the parameters are input and the steps for the selection of basis states are performed over a fixed basis dimension K. If a new basis parameter is found, an initial basis parameter is replaced. This is repeated cyclically. This step can be used to refine elements chosen in the two previous methods.

There are several methods for solving the generalized eigenvalue problem that does not involve its computation for each basis parameter.

- (1) Numerical integration over the radial $V(r)$ in order to calculate for $v(p)$ which contains all non-correlated Gaussian elements of the potential. For given values of p , this numerical integration is performed and $v(p)$ is tabulated. Values of $v(p)$ and p are given before the basis selection and are approximated using interpolation.
- (2) One can analytically solve for the potential.
- (3) Numerically integrate over the radial variable for each value of p that appears during the basis selection. This differs from the first method as the actual value of p is available only during the basis selection, resulting in this method being performed only during basis selection.

The methods presented above are able to be chosen freely from through the use of an input file for the Fortran code. Each method has its strengths and weaknesses. The methods are ordered by their speed with the first being the most swift.

Nonlinear elements of matrix A are necessarily generated randomly for each element. The first element is changed K_0 times and the best parameter for the lowest ground state energy is chosen to fill that spot. This process repeats for each diagonal element until each element is replaced. The entire process is repeated a total of M_0 times. The entire selection requires a total $N_0 \times M_0$ evaluations to complete, with the possibility of a fast evaluation due to only one element changing in each step.

Input data for the program is used to specify the details of the system which is being operated on. The number of particles, type of system, type of particles, spins, isospins, masses, and charges are all available to be changed. The system reads this data and then produces the ground state energy and wave function. Refining steps are then taken. This program is modifiable for a number of N particles determined by the capabilities of the computer being utilized.

Method Description:

The main equation to solve is the Eigenvalue equation

$$HC = E_K BC, \quad (4)$$

The smallest eigenvalue in this equation would be the ground-state energy for the system, which is the quantity we are looking for. For this, we will need to find the matrices H and B.

B is easy to calculate as it is merely the overlap of two basis functions.

$$B = \langle G_A | G_{A'} \rangle = \left(\frac{(2\pi)^{(N-1)}}{\det(A + A')} \right)^{\frac{1}{2}}$$

H is a bit harder, but since the Hamiltonian is a linear combination, it can be broken up into solvable parts.

$$H = \langle G_A | \mathcal{H} | G_{A'} \rangle = \langle G_A | \sum_{i=1}^N \frac{p_i^2}{2m_i} - T_{cm} | G_{A'} \rangle + \langle G_A | V_{ij} | G_{A'} \rangle$$

These terms can be solved using the processes described in the article.

By adapting the model to 1-dimensional, a couple of expressions need to be updated.

$$\langle G_A | \delta(r_i - r_j - r) | G_{A'} \rangle = \langle G_A | G_{A'} \rangle \left(\frac{1}{2\pi p_{ij}} \right)^{1/2} \exp \left(\frac{-r^2}{2p_{ij}} \right)$$

$$\nu(p_{ij}) = \left(\frac{1}{2\pi p_{ij}} \right)^{1/2} \int dr V(r) \exp \left(\frac{-r^2}{2p_{ij}} \right)$$

This allows us to find the ground-state energy for given parameters A_k^n and tune them to get lower values for the ground-state energy using the stochastic variational method.

Implementation:

In the implementation of the Stochastic Variational Method, we begin with the definition of the system. There are given input variables which may be freely changed. These variables are the number of particles (N), position of particles (r_i), masses (m_i), spins (s_i), isospins (t_i), and charge (z_i). Both the spins and isospins are represented by their projection along the z-axis (with up being coded as 1 and down being coded as 2). In addition, the potential of the system is to be defined arbitrarily with the default being a Gaussian potential defined in (2). This potential shall be incorporated such that it may be changed at will to another potential (Yukawa, Coulombic, or other numerical potentials). Other input data will include the definition of the particles as fermions or bosons (determined by a single numerical value with the variable set to one representing fermions while two represents bosons) and the ability to switch between three

methods of determining basis parameters (again represented by a numerical value for each of the three methods).

The input data mentioned above defines the few-body system which is being examined. However, input data is still needed for the process of random generation of basis parameters (which will be used in future steps). The values of $b_{\min}$ and $b_{\max}$ provide a range for the basis generation. M_0 provides the number of iterations desired for possible basis refinement while K_0 provides the maximum number of evaluations desired for the random parameters. The variable K determines the number of basis functions. A random seed will additionally be provided (and available for change) to aid in random number generation. The value of $\hbar^2/m$ will be defined as one for the purposes of most practical examples (but is available for modification).

After defining the input data, the Jacobi coordinates are calculated based on particle positions in a function. This is important as the Jacobi coordinates are translationally invariant which reduces possible complications with the center of mass motion. Next, the kinetic energy and potential energy functions are defined. Here is where the potential energy might be changed from a default Gaussian to a Yukawa, Coulombic, or harmonic potential. From there, the overlap function calculates the overlap matrix of the Gaussian basis function. The potential function serves to define how the system interacts and is where you may also choose the type of potential that the system experiences. The potential function models the interactions between the particles while the potential energy function models the potential energy in the system (determined by the type of potential).

The next step is the process of basis generation and refinement. This follows closely from the SVM where the process starts with choosing the method for determining parameters. The first method involves the pre-definition of bases (which would be included in input data). In this

case, it is best if the method of reading this data would be from reading a separate file from the local computer rather than modifying the input within the function itself. The first line of such a file would read the number of basis states K (although this may be easily alterable inside the code itself due to the defining of initial variables). The next lines would contain the non-linear parameters of the basis.

The second method involves the random generation of the parameters and the creation of the basis step-by-step through the addition of good parameters. This is performed by first generating several random sets of basis parameters. Then, by solving the energy eigenvalue problem, the parameter which corresponds with the lowest energy eigenvalue is chosen as the basis parameter. The parameter is then added to the previous basis states, with the basis dimension increasing by one. The only input data needed to perform this is the range over which basis parameters may be generated (defined by $b_{\min}$ and $b_{\max}$).

The third method involves the refinement of previously generated or defined bases. A basis is subjected to the random generation method described above over a fixed basis dimension K . If the parameter which is generated is better suited, the new parameter replaces the existing. This is repeated for each of the remaining parameters. The process is then repeated cyclically several times. This is perhaps best accomplished through a separate function which refines each parameter by searching in the neighborhood and analyzing whether the candidate energy is smaller than the current parameter's energy. If this is true, the candidate is chosen as the new parameter. In the implementation, all methods shall be available to be chosen from, although the program will be specifically designed to first generate random parameters and then refine them in the conjunction of methods two and three.

After the basis parameters are chosen, the generalized eigenvalue problem is solved. This should be accomplished numerically through a specific scientific library such as numpy or scipy where the eigenvalue problem might be solved by a function such as `np.linalg`. After the eigenvalues are found, the same function should attempt to find the minimum and then return the ground state energy. The simplifications defined in Reference [3] (a 1995 paper by Varga and Suzuki) determine that the entire eigenvalue problem need not be solved every time the basis changes. These simplifications will be implemented as well to avoid the significant resource sink of recalculating the energies for every change.

In a deeper examination of the potential function, it is possible to find this value through multiple ways. In the calculation of the value of $v(p)$ (which can be found in equation [17] from Varga and Suzuki), the function can be numerically integrated over the radial coordinate for some values of p . $v(p)$ can then be tabulated before the calculation even begins. During the generation basis functions, values of $v(p)$ at arbitrary values of p simply need to be interpolated for this to begin. This method is the fastest out of the methods mentioned and shall be the one we choose despite potential for inaccuracies. Using a dedicated numerical integration function (such as `scipy.integrate.quad`) the integral may be numerically evaluated. The values generated for $v(p)$ are then stored in an array. For the usage of interpolation techniques, we can use methods such as cubic splines interpolation (which we can obtain from scipy again). During this process, values of p should be sourced close together in order to ensure the accuracy of interpolation later. Error control might additionally be implemented through the checking of the interpolated value against the exact value calculated numerically and adjusting the tabulated grid as needed.

Due to the complexity and the number of these functions, an overarching main function will be established whose goal is to orchestrate and coordinate the flow of the program from the

inputs, basic functions, basis parameters, and eigenvalue problems to the output. The establishment of this function will also allow for easy modification of the input variables mentioned as system-defining. The outputs produced by this function would be the ground state energy values (specifically upper bounds on ground state energy values) and the wave function.

Group Assessment:

The division of labor for this project was evenly distributed as both members of the group sought to advance their conceptual and mathematical understanding of the material jointly.

Brendan Santoianni completed the summary and the implementation plan while Aniruddh Nukal sought to produce the method description and analysis for the transition from three dimensions to one dimensions for the matrix elements. In all parts of the assignment, both group members collaborated and offered advice.